Data Sheet

EXPERIMENT NO. 03

Title: - Simulation of Transistor Circuit with Fixed-Bias Configuration and Study of Output Parameters (using Tinkercad/Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity

Important Formulae: -

Base Current (I_B): -

$I_B = (V_{CC} - V_{BE}) / R_B$

- V_{CC} = Supply voltage
- V_{BE} = Base-emitter voltage (0.7 V for Si transistor, 0.3 V for Ge transistor)
- R_B = Base resistor

Collector Current (I_C): -

$I_C = \beta \times I_B$

- β = Current gain (transistor parameter)

Collector-Emitter Voltage (V_{CE}) — Theoretical: -

$V_{CE} = V_{CC} - I_C \times R_C$

Stability Factor: -

$S = (\beta + 1)$

Observation Table: -

Sl. No.	I_B (µA)	I_C (mA)	R_C (Ω)	V_{CE} (Practical) (V)	V_{CE} (Theoretical) (V)

Sl. No.	I_B (µA)	I_C (mA)	R_C (Ω)	V_{CE} (Practical) (V)	V_{CE} (Theoretical) (V)

Calculation: -

I_B = _____

$I_C = \beta \times I_B$ = _____

V_{CE} (Theoretical) = $V_{CC} - I_C \times R_C$ = _____

% Error between Practical and Theoretical $V_{CE} = [(V_{CE(Th)} - V_{CE(Pr)}) / V_{CE(Th)}] \times 100$ = _____ %

Teacher's Sign and Date

Data Sheet

EXPERIMENT NO. 04

Title: - Simulation of Transistor Amplifier with Potential Divider Bias to Study Output Voltage and Gain for an Input Signal (using Tinkercad/proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity

Important Formulae: -

Base Voltage (VB): -

$V_B = V_{CC} \times [R_2 / (R_1 + R_2)]$

Emitter Voltage (VE): -

$V_E = V_B - V_{BE}$

- VBE = 0.7 V (for Si transistor)

Emitter Current (IE) and Collector Current (IC): -

$I_E = V_E / R_E$ and $I_C \approx I_E$

Collector-Emitter Voltage (VCE) — Theoretical: -

$V_{CE} = V_{CC} - I_C \times (R_C + R_E)$

Voltage Gain (Av): -

$A_V = V_{out} / V_{in}$

Observation Table: -

Sl. No.	I_B	I_C	R_1	R_2	R_C	R_E	V_{CE} (Practical)	V_{CE} (Theoretical)

Sl. No.	I_B	I_C	R_1	R_2	R_C	R_E	V_{CE} (Practical)	V_{CE} (Theoretical)

Gain Observation Table: -

Sl. No.	Vin (mV)	Vout (V)	Voltage Gain, Av = Vout/Vin

Calculation: -

$V_B = V_{CC} \times [R_2/(R_1+R_2)]$ = _____

$V_E = V_B - V_{BE}$ = _____

$I_E = V_E / R_E$ = _____ , $I_C \approx I_E$ = _____

V_{CE} (Theoretical) = $V_{CC} - I_C(R_C + R_E)$ = _____

Voltage Gain, Av = Vout / Vin = _____

Teacher's Sign and Date